Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/02/2013
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08544	**Congressional District:**	12
Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2009	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Project extended to 9/30/2015, per D. Stocker/GRC and NSSC, pending award of new grant (Ed., 10/6/14) NOTE (Ed., 8/5/2014): This project is being continued as a new grant to be awarded in fall 2014. See that task for subsequent reporting. NOTE: Project extended to 9/30/2014 per NSSC information/D.Stocker-GRC (Ed., 8/2012) NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen (Rutgers University)		
Grant/Contract No.:	NNX09AR03A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	FY2014 ANNUAL REPORT The extent of rigor in the various claims of self-similarity in previous studies of the self-acceleration of expanding spherical flames was examined, particularly on the attainment of 3/2 for the power-law acceleration exponent. We find that the evidences presented in some of the prior studies were not rigorously interpreted, and as such the claim of the 3/2 exponent, motivated by the initial suggestion of Gostintsev et al. based on results of uncertain accuracy, and by the attractiveness of its potential implication on self-turbulization, is not justified. Our results suggest that self-acceleration definitely exists; and self-similarity, subsequent to the initial transient, most likely exists. Our power-law exponent, however, still seems to weakly depend on the system pressure and the intensity of the diffusional-thermal instability, and as such has not reached the hydrodynamic, flame-sheet limit. Further experiments with even thinner flames and longer observation times are needed to resolve this scientifically and practically important issue on flame propagation.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Wu F, Jomaas G, Law CK. "An experimental investigation on self-acceleration of cellular spherical flames." Proceedings of the Combustion Institute. 2013;34(1):937-45. http://dx.doi.org/10.1016/j.proci.2012.05.068 , Jan-2013
Papers from Meeting Proceedings	Lang W, Law CK, Chen Z. "Role of Soret Diffusion in the Ignition of Hydrogen/Air Mixtures by a Heated Kernel." 8th U. S. National Combustion Meeting, The Combustion Institute, Park City, Utah, May 19-22, 2013. 8th US National Combustion Meeting 2013. Proceedings of a meeting held 19-22 May 2013, Park City, Utah, USA. Livermore, Calif. : Western States Section/Combustion Institute, 2013. p. 783-793. Paper # 070LT-0030. Printed by Curran Associates, Inc. ISBN: 9781627488426. , Nov-2013